

Name: K RAJ BHARATH K RAJ BHARATH | Reg. No: RA2511003011548

## Unit – 5 : Session – 4 : SLO – 1 | Parallelism, Control Unit and Pipelining Review

### ■ I. Multiple Choice

1. b) Executing multiple instructions simultaneously within a single processor.
2. c) Coordinate and control the operations of other units.
3. c) Overlapping multiple instruction stages.
4. c) Synchronization and communication overhead.
5. b) Memory consistency problems — shared data cached per-core must stay coherent/consistent.

### ■ II. True or False

1. FALSE — Amdahl's Law: the serial fraction and overheads prevent linear speedup.
2. TRUE.
3. TRUE.
4. FALSE — communication overhead increases with more processors.
5. TRUE.

### ■ III. Scenario-Based Activity

1. Idle/overloaded cores = load imbalance → solution: dynamic work scheduling (work stealing, finer-grained tasks).
2. Poor scaling with more cores = Amdahl/serial fraction + synchronization overhead → solution: reduce the serial portion, minimize lock scope, use lock-free structures.
3. Results differ between runs = memory consistency/race conditions → solution: proper synchronization (locks, atomics, barriers) or a defined consistency discipline.
4. Waiting for data updates = communication overhead → solution: data locality, partition work to minimize sharing, overlap communication with computation.